

CAFÉ FINDER

Rash

1. Introduction

With the rapid growth of mobile technology and location-based services, people increasingly rely on digital applications to discover nearby places and services. One of the most common needs, especially among students, professionals, and travelers, is finding suitable cafes based on their preferences such as location, price range, ambiance, and available facilities.

Popular platforms like Zomato, Yelp, and Google Maps already provide solutions for locating restaurants and cafes. However, these applications are often overloaded with features, making it difficult for users who are specifically searching for cafes to quickly find relevant results.

The proposed system, **CAFE FINDER**, is a mobile-based application designed to help users easily discover nearby cafes based on their current location and preferences. The system focuses specifically on cafes, providing a simplified and user-friendly interface while delivering essential information such as location, ratings, reviews, and available facilities.

The aim of this application is to provide a fast, efficient, and personalized experience for users who are looking for cafes in unfamiliar or familiar areas.

2. Background

In today's fast-paced lifestyle, cafes have become more than just places to eat and drink. They serve as social hubs, study environments, workspaces, and relaxation spots. Especially in urban areas, users often struggle to identify suitable cafes that meet their specific needs, such as availability of Wi-Fi, quiet environments for studying, or affordable pricing.

Existing applications such as Zomato and Yelp provide extensive listings of restaurants, but they lack specialization in cafe-related features. Additionally, users may experience challenges such as:

- Information overload due to too many options
- Lack of personalized recommendations
- Difficulty in filtering cafes based on specific needs (e.g., study-friendly, less crowded)
- Inconsistent or outdated user reviews

Furthermore, although Google Maps provides location-based search and navigation, it is a general-purpose application and does not focus on enhancing the cafe discovery experience.

To address these limitations, the **CAFE FINDER** system is proposed as a dedicated solution that focuses solely on cafe discovery. The application will utilize location-based services (GPS) to identify nearby cafes and present them in an organized and user-friendly manner. It will also incorporate features such as filtering options, user reviews, and personalized recommendations to improve user experience.

By narrowing the scope to cafes and enhancing usability, the system aims to provide a more efficient and targeted solution compared to existing general-purpose applications.

3. Problem Statement

In today's digital era, users frequently depend on mobile applications to locate nearby cafes and dining options. However, most existing platforms such as Zomato, Yelp, and Google Maps are designed for general restaurant discovery rather than focusing specifically on cafes.

As a result, users face several challenges:

- Difficulty in quickly finding **cafes that match specific needs** (e.g., study-friendly, Wi-Fi availability)
- **Information overload** due to excessive listings and features
- Lack of **personalized recommendations**
- Limited filtering options tailored specifically for cafes
- Inconsistent or outdated reviews

These limitations create a need for a **dedicated, user-friendly system** that focuses exclusively on cafe discovery and enhances the overall user experience.

4. Project Objectives

Main Objective

To develop a user-friendly mobile/web-based application (**CAFE FINDER**) that helps users easily discover and explore nearby cafes based on their preferences and location.

Specific Objectives

- To provide **location-based cafe search** using GPS
 - To enable **filtering options** (price, rating, facilities, distance)
 - To allow users to **view cafe details** (menu, images, reviews)
 - To implement a **user rating and review system**
 - To provide **personalized cafe recommendations**
 - To design a **simple and intuitive user interface**
 - To support users in finding cafes suitable for **studying, working, or relaxing**
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5. Introduction to Similar Type of Systems / International Examples

Several international applications provide location-based cafe and restaurant discovery services:

- Zomato
Offers restaurant listings, menus, ratings, and online ordering features.
- Yelp
Focuses on user reviews, ratings, and advanced filtering options.
- Google Maps
Provides map-based search, directions, and popular place recommendations.
- Foursquare
Suggests places based on user preferences and past activity.

Although these systems are powerful, they are **not specialized for cafes**, which reduces efficiency for users specifically searching for cafe-related experiences.

6. Sri Lankan Cafe Finder Websites & Apps

In Sri Lanka, there is currently **no dedicated cafe-only finder platform**, but several websites and mobile applications provide **restaurant and food discovery services** that indirectly help users find cafes. These systems combine location-based services, reviews, and listings to guide users.

1. Melo

- A Sri Lankan mobile app designed to help users discover restaurants and dine-in places
- Provides:
 - Location-based search
 - Menu browsing
 - User ratings and reviews
- Includes innovative features like **3D food visualization**
- Covers multiple food categories including cafes

Relevance to Cafe Finder:

Users can search for cafes, but the app is **not specialized** for cafe-specific needs.

2. Kohedha

- A local discovery platform focused on:
 - Food places
 - Events
 - Trending venues
- Allows filtering by:
 - Location
 - Budget
 - Vibe (romantic, rooftop, casual, etc.)
- Provides **nearby recommendations and trending places**

👉 **Relevance to Cafe Finder:**

Strong in **experience-based discovery**, but still not focused only on cafes.

3. TryThisFood

- A community-driven platform for:
 - Tracking food experiences
 - Posting reviews and photos
- Features:
 - Location-based discovery
 - AI-powered hashtags
 - User-generated content

👉 **Relevance to Cafe Finder:**

Good for **reviews and social interaction**, but lacks structured cafe filtering.

4. QPON

- Focuses on:
 - Discounts and offers for cafes and restaurants
 - Real-time location-based deals
- Helps users discover **hidden local spots and cafes nearby**

👉 **Relevance to Cafe Finder:**

Useful for finding cafes with **offers**, but not for detailed exploration.

5. JustEatLanka

- A food delivery platform that allows users to:
 - Find nearby restaurants and cafes

- Order food online
- Includes **location-based browsing and payment options**

☞ **Relevance to Cafe Finder:**

Primarily focused on **delivery**, not cafe discovery experience.

6. Java Lounge App

- Dedicated app for a **specific cafe chain** in Sri Lanka
- Features:
 - Online ordering
 - Loyalty rewards
 - Menu browsing

☞ **Relevance to Cafe Finder:**

Limited to **one brand**, not useful for discovering multiple cafes.

However, these platforms:

- Are **not focused specifically on cafes**
- Lack **custom filtering for cafe-specific needs**
- Do not provide a **localized personalized experience** for Sri Lankan users

7. Identified Gaps in Existing Systems

After analyzing existing systems, the following gaps were identified:

- No **dedicated platform** for cafe discovery
- Lack of **filters for study/work-friendly cafes**
- Limited support for **real-time crowd or availability information**
- Poor **user experience due to complex interfaces**
- Lack of **localized recommendations** for Sri Lankan users
- Minimal focus on **user personalization and preferences**

8. Proposed Solution

To overcome the identified gaps, the proposed solution is the development of **CAFE FINDER**, a dedicated cafe discovery application.

The system will:

- Focus **only on cafes** to reduce complexity
- Provide **accurate location-based results**

- Offer **custom filters and personalized suggestions**
- Ensure a **simple, fast, and user-friendly interface**
- Include **user-generated content** such as reviews and ratings

This solution aims to deliver a more **efficient and targeted experience** compared to existing applications.

9. Key Features of the Proposed System

-  Location-based cafe search
-  Advanced filtering (price, rating, facilities)
-  Map integration with directions
-  User ratings and reviews
-  Favorite/save cafes option
-  Cafe images and menus
-  Personalized recommendations
-  Filters for Wi-Fi, charging, study environment

10. Technology Planning

The system will be developed using modern web/mobile technologies to ensure performance, scalability, and usability.

Recommended Tech Stack for Cafe Finder App

Frontend (Mobile App)

☒ Best Choice:

- **React Native**
- Language: **JavaScript (or TypeScript if you want bonus points)**

Why this is a smart choice:

- Build **Android + iOS apps with one codebase**
- Huge demand in internships/jobs
- Works perfectly with Node.js backend
- Reusable React knowledge (web + mobile)

What you'll use:

- **React Native components (UI)**

- Navigation (React Navigation)
 - API calls (Axios / Fetch)
 - Maps (Google Maps API)
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Backend (Server Side)

☒ Best Choice:

- **Node.js**
- Framework: **Express.js**

Why:

- Same language (JavaScript) → easier learning
- Very popular in internships
- Fast and scalable
- Perfect for REST APIs

What it handles:

- User authentication (login/signup)
 - Cafe data management
 - API for frontend
 - Recommendation logic
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Database

☒ Best Choice:

- **MongoDB**

Why:

- Flexible (good for cafe data like menus, images, reviews)
- Works smoothly with Node.js
- Easy to learn for beginners
- Widely used in modern apps

Alternative (optional):

- MySQL → if your course requires relational DB

Full Stack Summary (What YOU should use)

🔑 This is your final answer (you can copy to report):

- **Frontend:** React Native (JavaScript)
- **Backend:** Node.js with Express.js
- **Database:** MongoDB
- **API Type:** REST API

Why this stack is PERFECT for internships

Right now, companies look for:

- JavaScript developers
- React / React Native developers
- Node.js backend developers

🔑 By using this stack, you can say:

“I developed a full-stack mobile application using React Native, Node.js, and MongoDB”

That line alone = 💧 for your CV

Bonus (if you want to stand out more)

If you want to go one level higher:

- Use **TypeScript** instead of plain JS
 - Add **Firebase Auth** (for login)
 - Deploy backend (Render / Railway)
 - Use GitHub properly (very important!)
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🎯 Realistic Advice (important)

Don't try to learn everything at once. Start like this:

1. React Native basics
2. Simple UI (list of cafes)
3. Node.js API
4. Connect app → backend
5. Add database

Short answer

👉 React Native is NOT a programming language

👉 JavaScript is the programming language

React Native = a framework

JavaScript = the language you write code in

📱 What exactly is React Native?

React Native is a framework developed by Meta that lets you build **mobile apps (Android + iOS)** using JavaScript.

So instead of writing:

- Java (for Android)
- Swift (for iOS)

👉 You write **JavaScript once**, and React Native turns it into a real mobile app.

□ How React Native and JavaScript work together

Think of it like this:

- **JavaScript** → the language (like English)
- **React Native** → the tool/framework (like a translator that turns your words into a mobile app)

Example:

```
import { Text, View } from 'react-native';

export default function App() {
  return (
    <View>
      <Text>Hello Cafe Finder</Text>
    </View>
  );
}
```

👉 This is **JavaScript code**, but using **React Native components** (View, Text)

🔄 React vs React Native (quick clarity)

You might also hear about React, so here's the difference:

- **React**
 - Used for **websites**
 - Uses HTML (div, h1, etc.)
- **React Native**
 - Used for **mobile apps**
 - Uses mobile components (View, Text)

👉 Both use **JavaScript + same concepts**

💡 What you need to learn first

Since you haven't learned React Native yet, follow this order:

1 JavaScript (VERY IMPORTANT)

Learn basics:

- Variables
- Functions
- Arrays & objects
- ES6 (arrow functions, imports)

2 React basics

- Components
- Props

- State

3 React Native

- UI components (View, Text, Image)
- Navigation
- API calls

🌀 Simple conclusion (you can remember this)

👉 JavaScript = language

👉 React Native = framework that uses JavaScript

They are **not separate languages**, but they are **not the same thing either**

Conclusion

The development of the **CAFE FINDER** system addresses a clear gap in both international and Sri Lankan digital platforms related to cafe discovery. While popular applications such as Zomato, Yelp, and Google Maps provide general location-based services, they do not specifically focus on the unique requirements of cafe users.

Through the analysis of existing systems, it is evident that users often experience challenges such as information overload, lack of personalized recommendations, and difficulty in identifying cafes that meet specific needs such as a suitable study environment or availability of facilities like Wi-Fi.

In the Sri Lankan context, current platforms also lack a dedicated solution for cafe discovery, further emphasizing the need for a specialized system. The proposed **CAFE FINDER** application aims to overcome these limitations by offering a focused, user-friendly, and efficient platform tailored specifically for cafe exploration.

By utilizing modern technologies such as React Native, Node.js, and MongoDB, the system ensures scalability, performance, and ease of use. Additionally, the inclusion of features such as location-based search, advanced filtering, user reviews, and personalized recommendations enhances the overall user experience.

In conclusion, the **CAFE FINDER** system provides a practical and innovative solution that not only improves cafe discovery but also aligns with current technological trends and user expectations. This project has the potential to evolve into a valuable application for both local users and tourists, contributing positively to the digital ecosystem of Sri Lanka.

